

Vivek Chandrakant Bhave

bhavevivek7@gmail.com +91-9325933710 LinkedIn GitHub

Education

M.S. in Artificial Intelligence & Data Science

2025 – 2027

Atal Bihari Vajpayee Indian Institute of Information Technology and Management (ABV-IIITM), Gwalior
CGPA (Semester 1): 9.32

Skills

Programming: C, C++, Python, SQL

Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

Machine Learning: Regression, Classification, Clustering, Model Evaluation

Data Analysis: EDA, Data Cleaning, Feature Engineering, Data Visualization

Tools: Git, GitHub, Jupyter Notebook, VS Code

Projects

IMDB Sentiment Analysis Pipeline

Python, Scikit-learn, TF-IDF, Pandas, SQL, Git

- Built an end-to-end sentiment analysis pipeline on **40k IMDB reviews** including EDA, TF-IDF feature engineering, and classical ML models.
- Tuned TF-IDF parameters (**min_df**, **max_df**, **max_features**, **n-grams**) to reduce noise and improve classification accuracy.
- Trained and compared multiple models: **Logistic Regression (0.93)**, **SVM (0.89)**, **Naive Bayes (0.87)**, **Random Forest (0.85)**.
- Performed detailed metric analysis using **precision**, **recall**, **F1-score**, and **confusion matrix** to evaluate real-world reliability instead of accuracy only.
- Designed the project for real-world use cases such as **large-scale review monitoring and audience feedback analysis**; failure analysis of misclassified reviews in progress.
- GitHub: imdb-sentiment-analysis-pipeline

Human-Relatedness Embedding

Python, GloVe, NumPy, Pandas, Scikit-learn

- Evaluated how well word embeddings align with human semantic judgments using **SimLex-999** and **MEN** datasets.
- Computed cosine similarity across **GloVe embeddings (50D–300D)** and measured alignment using **Pearson correlation** and **R²**.
- Achieved best results on MEN dataset: **correlation = 0.74** and **R² = 0.55** (300D), showing strong relatedness alignment.
- GitHub: Human-Relatedness-embedding

Achievements

- Practicing Data Structures Algorithms: solved 38 problems on LeetCode (Arrays, Strings, Sliding Window, Prefix Sum, Recursion, Binary Search) — LeetCode Profile
- Maintained a public DSA learning repository in C++ — GitHub